Biologically-inspired adaptive learning in the Hopfield-network based self-optimization model

NEURAL INFORMATION PROCESSING SYSTEMS

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Paper **Join me on Zoom!**

Biological learning is different from machine learning in key ways

Biological systems operate with:
- sparse data
- lower/efficient energy use
- minimal/variable training
- trade-offs **driving adaptation**

Machine learning operates with:
- large datasets
- high energy use
- extensive training
- trade-offs causing **failure**

What **biological mechanisms** could help **enhance** ML?

Self-optimization (SO) model

The SO model is a Hopfield network with:
- updating of weights based on a Hebbian learning rule
- periodic resets of network states

Able to **combine optimization** and memory **stabilization**?

Example: learning rate as a function of energy

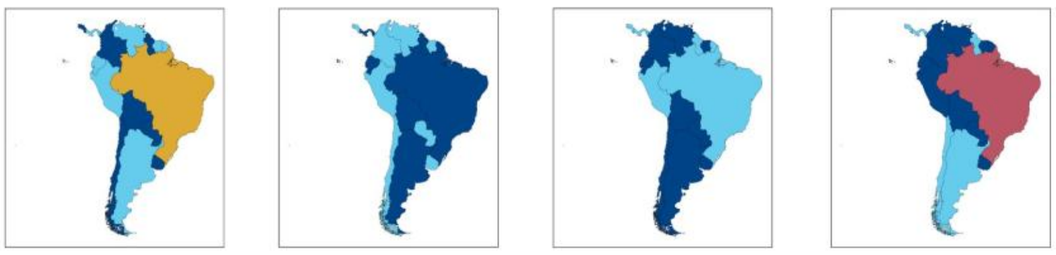
1. Weight update rule (α: learning rate; s_i, s_j: states of nodes i and j)

$$w_{ij}(t+1) = w_{ij}(t) + \mathrm{d}w_{ij}(t), \quad \mathrm{d}w_{ij} = \alpha s_i s_j$$

2. Learning rate α as a function of energy E:

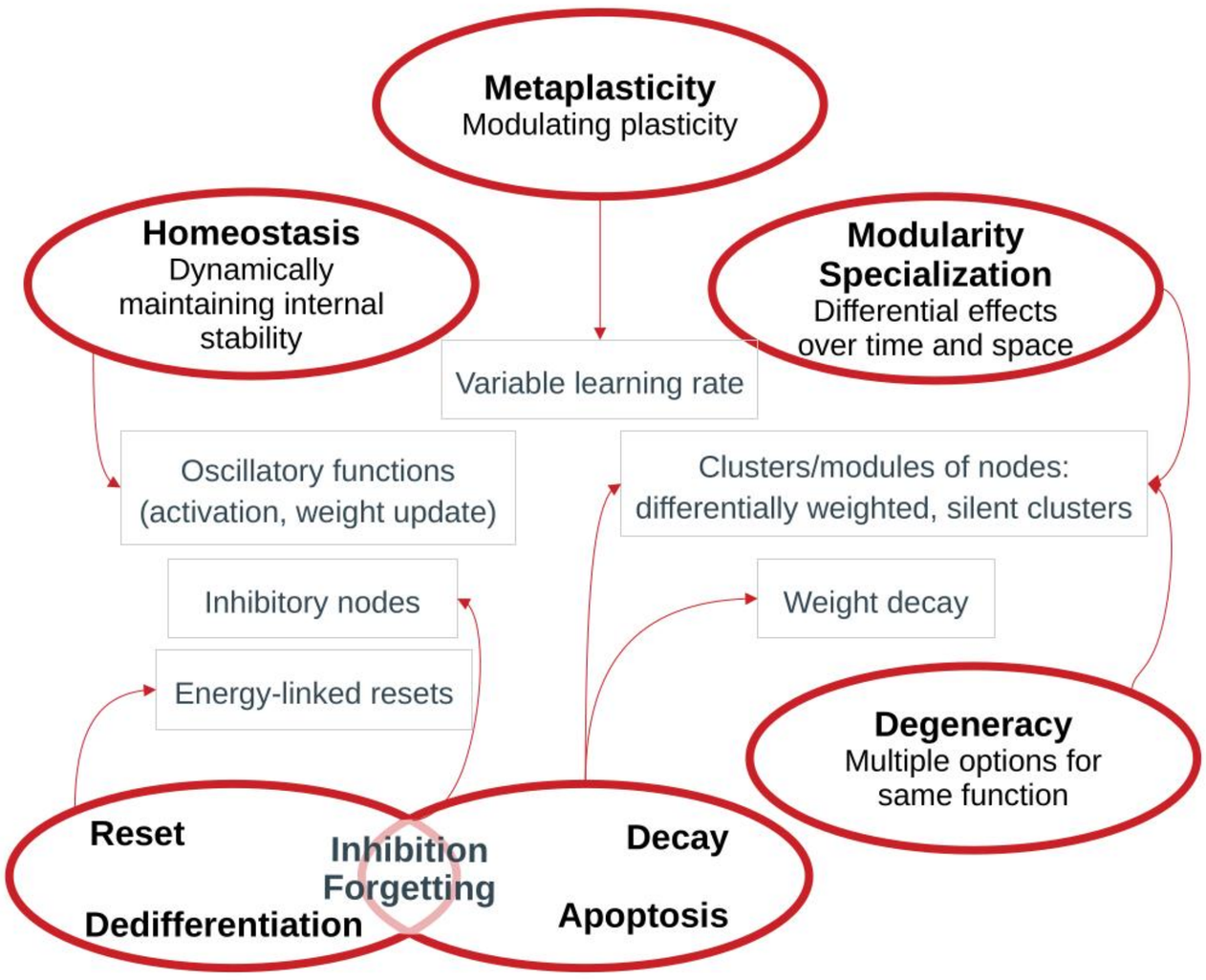
$$\alpha = \begin{cases} \frac{a}{\Delta E + \epsilon_0} & \gamma \neq 1 \\ \alpha_0 & \gamma = 1 \end{cases} \quad \text{with} \quad \epsilon_0 = \frac{\gamma}{1-\gamma}, \quad a = \alpha_0 \epsilon_0 = \alpha_0 \frac{\gamma}{1-\gamma}$$

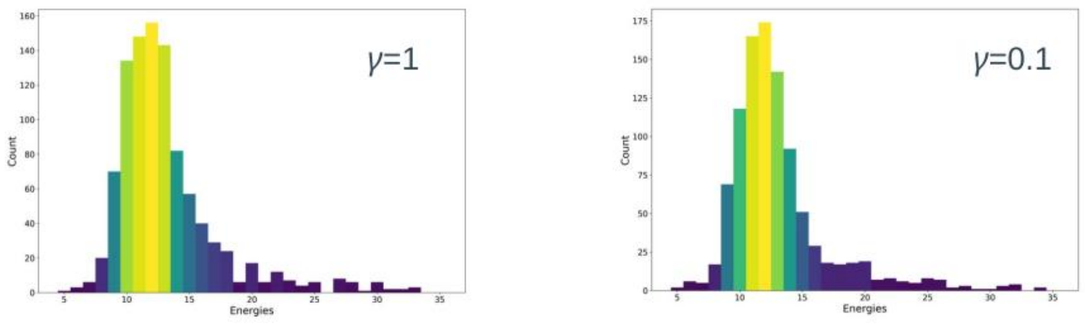
SO model results for map-coloring problem. From left to right, E=5, E=8, E=12, E=20 (only the two middle results are valid maps).



Implementing mechanisms used by living systems to adapt to their environment may help solve trade-offs inherent to machine learning.

Translating adaptive biological mechanisms into a Hopfield network





Energy distributions at the end of learning for 1000 runs. A slight difference can be observed between γ=1 (learning rate α constant) and γ=0.1 (variable learning rate), but not obviously significant.

What next?

- Test further implementations of variable learning rate: as function of weights, trigonometric function
- Test "forgetting": weight decay, inhibition
- Applications in continuous learning, creativity, robotics
- Modeling biological systems

References

W. C. Abraham. Metaplasticity: Tuning synapses and networks for plasticity. Nat Rev Neurosci, 9(5):387–387, 2008-05

H. Kitano. Biological robustness. Nat Rev Genet, 5(11):826–837, 2004-11

R. A. Watson, C. L. Buckley, and R. Mills. Optimization in "self-modeling" complex adaptive systems. Complexity, 16(5):17–26, 2011

N. Weber, W. Koch, O. Erdem, and T. Froese. On the use of associative memory in Hopfield networks designed to solve propositional satisfiability problems, 2023-08-14

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